Ahmad Rahimi

Department of Computer Science
Ecole Polytechnique Federale Lausanne (EPFL)
Lausanne, Switzerland

ahmad.rahimi@epfl.ch
ahmadrahimiuni@gmail.com
Cell Phone: +41762669437
Homepage

Summary

Researcher in video generation, driving world models, autonomous driving, diffusion-based generation, 4D scene reconstruction, and trajectory prediction. Published at CVPR 22, 25, and 26, and ICML 26. Interned at Wayve on world-modeling, VR-based driving-scene analysis, and 4D scene reconstruction.

Education

- **Ph.D. Degree in Computer Science** — Sept. 2022 – present
  Department of Computer Science — Supervisor: Alexandre Alahi
  École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland

- **B.Sc Degree in Computer Science** — Sept. 2018 – 2022
  Department of Mathematical Sciences — **GPA: 3.94/4 (18.88/20)**
  Sharif University of Technology, Tehran, Iran

Work Experience

- **Research Intern at Wayve** — Nov. 2025 - May 2026
  Worked with the GAIA team on driving world models and data-centric evaluation.
  - Built a pipeline for generating 360° panoramic videos from recorded driving scenarios, together with a Meta Quest based VR review application placing users inside a Wayve vehicle environment with panoramic video playing; improved labeling quality and increased intervention-retrieval recall by over 3x.
  - Designed feed-forward and recursive-refinement architectures for explicit 4D driving-scene reconstruction and generation with dynamic 3D Gaussian splats, using rendered intermediate predictions and conditioning images to iteratively improve scene estimates.

Selected Publications

- **"MAD: Motion Appearance Decoupling for efficient Driving World Models"**,
  Ahmad Rahimi*, Valentin Gerard*, Eloi Zablocki, Matthieu Cord, and Alexandre Alahi
  Introduces a two-stage driving world model that decouples motion forecasting from appearance synthesis, enabling efficient video generation with only 140 GPU hours of training, two orders of magnitude less than prior work. **(CVPR 2026)**, project page

- **"GEM: A Generalizable Ego-vision Multimodal World Model for Fine-Grained Ego-Motion, Object Dynamics, and Scene Composition Control"**,
  Mariam Hassan*, Sebastian Stapf*, **Ahmad Rahimi***, and 17 more authors.
  Develops a controllable ego-vision world model for long-horizon driving-video generation using multimodal controls over ego-motion, object dynamics, and scene composition, built on a Stable Video Diffusion backbone. **(CVPR 2025)**, link to arxiv

- **"Sim-to-Real Causal Transfer: A Metric Learning Approach to Causally-Aware Interaction Representations"**,
  Ahmad Rahimi*, Po-Chien Luan*, Yuejiang Liu*, Frano Rajič, and Alexandre Alahi.
  Learns causally-aware interaction representations for multi-agent trajectory prediction, improving robustness under out-of-distribution and low-data settings. **(CVPR 2025)**, link to arxiv

OTHER PUBLICATIONS

- **"Vehicle trajectory prediction works, but not everywhere"**, Mohammadhossein Bahari*, Saeed Saadatnejad*, **Ahmad Rahimi**, Mohammad Shahverdi Kondori, Seyed-Mohsen Moosavi-Dezfooli, and Alexandre Alahi. **(CVPR 2022)**, link to arxiv

- **From Generation to Generalization: Emergent Few-Shot Learning in Video Diffusion Models** Pablo Acuaviva, Aram Davtyan, Mariam Hassan, Sebastian Stapf, **Ahmad Rahimi**, Alexandre Alahi, and Paolo Favaro. **(ICML 2026)**, link to arxiv

- **"A Multi-Loss Strategy for Vehicle Trajectory Prediction: Combining Off-Road, Diversity, and Directional Consistency Losses"**, **Ahmad Rahimi** and Alexandre Alahi. link to arxiv

HONORS AND AWARDS

- **Third Place in LLM Hackathon at EPFL** — Apr. 2024
  a 2 days hackathon on improving LLM architectures like GPT 2 and Llama 2 held at EPFL.
- **Bronze Medal** — Sept. 2017
  in the Iranian National Olympiad in Informatics (INOI).
- **Ranked Second in !Optimizer Contest** — Sept. 2021
  a nation-wide scientific contest in optimization held in Sharif University of Technology
- **Ranked Third in Webelopers Contest** — Nov. 2019
  a contest for web developers in Django among students in Sharif University of Technology
- **Ranked 403** — Sept. 2018
  in the nationwide universities entrance exam among more than 200,000 participants in Iran (placed in top 0.2% of the participants)

TEACHING EXPERIENCE

- **Teaching Assistant, EPFL** — 2023–2024
  Deep Learning for Autonomous Vehicles; Introduction to Programming.
- **Teaching Assistant, Sharif University of Technology** — 2019–2021
  Computer Vision; Image Processing; Combinatorial Optimization; Probability; Programming.
- **Olympiad in Informatics Teacher, Allameh Helli 10** — June 2018 – Sept. 2018
  Algorithm design and data structures.

LEADERSHIP AND SERVICE

- **Head of Iranian Student Association, EPFL** — Aug. 2023
  Led cultural and scientific events, including a Nowruz event with 250+ participants.
- **Head of Student Scientific Association, Sharif University of Technology** — June 2019
  Led student scientific activities in the Department of Mathematical Sciences.
- **Co-organizer, Sharif Mathematical Sciences Summer School** — Aug. 2019
  Organized a summer school for 80 high-school students covering mathematics and computer science topics.

SKILLS

- **Vision and Gen-AI**: Video Generation, World Models, Diffusion Models, 4D Gaussian Splats
- **Deep Learning**: PyTorch (expert), TensorFlow (experienced), NumPy (expert), JAX (familiar)
- **Programming Languages**: Python (expert), Java (experienced), C++ (experienced)
- **Document Preparation**: LaTeX (expert), Office (expert)
- **Tools**: Git (expert), Docker (experienced), WandB (expert)

LANGUAGES

- Persian (native)
- English (fluent, TOEFL:[Total:110, Reading:30, Listening:30, Speaking:24, Writing:26])
- French (B1, intermediate)